

Partial correlation between poverty rate and number of crimes committed given literacy rate

```
# Loading the data set:-
```

```
x = read.delim("C:/Users/rahul/OneDrive/Documents/ALL.txt")
```

```
# Knowing some primary things about the data set:-
```

```
dim(x)
```

```
## [1] 34 4
```

```
names(x)
```

```
## [1] "State.UT"
```

```
## [2] "Total.Crime.against.Women..IPC..SLL."
```

```
## [3] "Literacy.rate"
```

```
## [4] "Poverty.rate"
```

```
# Computing Partial correlation:-
```

```
f3 = lm(Total.Crime.against.Women..IPC..SLL.~Literacy.rate , data = x) # Regress Total.Crime.against.Women..IPC..SLL. on Literacy.rate
```

```
f4 = lm(Poverty.rate~Literacy.rate , data = x) # Regress Poverty.rate on Literacy.rate
```

```
f13 = f3$residuals # Taking residuals(true value - estimated value) of f3
```

```
f14 = f4$residuals # Taking residuals(true value - estimated value) of f4
```

```
r = cor(f13,f14) # Partial correlation between poverty rate and number of crimes committed
```

```
r # Print required partial correlation
```

```
## [1] 0.2116541
```